

Program 7: Deploy a web application to Kubernetes using Minikube

Step 1: Start the minikube and check status

```
(base) 1rv24mc043_Honeyc@Yoga-7-2-in-1-141ML9:~$ mkdir p7
(base) 1rv24mc043_Honeyc@Yoga-7-2-in-1-141ML9:~$ cd p7
(base) 1rv24mc043_Honeyc@Yoga-7-2-in-1-141ML9:~/p7$ minikube start
minikube v1.37.0 (Ubuntu 22.04)
🔧 Using the docker driver based on existing profile
👉 Starting "minikube" primary control-plane node in "minikube" cluster
🐳 Pulling base image v0.0.48 ...
🔄 Restarting existing docker container for "minikube" ...

Docker is nearly out of disk space, which may cause deployments to fail! (96% of capacity). You can pass '--force' to skip this check.
💡 Suggestion:

Try one or more of the following to free up space on the device:

1. Run "docker system prune" to remove unused Docker data (optionally with "-a")
2. Increase the storage allocated to Docker for Desktop by clicking on:
   Docker icon > Preferences > Resources > Disk Image Size
3. Run "minikube ssh docker system prune" if using the Docker container runtime
🔗 Related issue: https://github.com/kubernetes/minikube/issues/9024

🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   • Using image gcr.io/k8s-minikube/storage-provisioner:v5
   • Using image registry.k8s.io/metrics-server/metrics-server:v0.8.0
   • Using image registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.2
   • Using image registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.2
   • Using image registry.k8s.io/ingress-nginx/controller:v1.13.2
🔍 Verifying ingress addon...
🔧 Enabled addons: metrics-server, storage-provisioner, default-storageclass, ingress

! /usr/local/bin/kubectrl is version 1.31.0, which may have incompatibilities with Kubernetes 1.34.0.
• Want kubectrl v1.34.0? Try 'minikube kubectrl -- get pods -A'
👉 Done! kubectrl is now configured to use "minikube" cluster and "default" namespace by default
```

Step 2: Check status of minikube

```
(base) 1rv24mc045_Honey@Yoga-7-2-in-1-14IML9:~/p7$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

Step 3: Create a nginx-deployment file

nano nginx-deployment.yaml

```
GNU nano 7.2 nginx-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80
```

Step 4: Apply the yaml created

Step 5: Check for the deployments and pods

```
(base) 1rv24mc043_Honey@Yoga-7-2-in-1-14IMLR:~/p7$ nano nginx-deployment.yaml
(base) 1rv24mc043_Honey@Yoga-7-2-in-1-14IMLR:~/p7$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created
(base) 1rv24mc043_Honey@Yoga-7-2-in-1-14IMLR:~/p7$ kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
```

Step 6: Expose the deployment

Step 7: Get the services to check for deployments

```
(base) 1rv24mc043_Honey@Yoga-7-2-in-1-14IML9:~/p7$ kubectl expose deployment nginx-deployment --type=NodePort --port=80
service/nginx-deployment exposed
(base) 1rv24mc043_Honey@Yoga-7-2-in-1-14IML9:~/p7$ kubectl get services
NAME                TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes          ClusterIP   10.96.0.1    <none>        443/TCP          11h
nginx-deployment    NodePort    10.100.101.145 <none>        80:31281/TCP     7s
```

Step 8: Get the nginx official page by using the below command in the browser

```
(base) 1rv24mc043_Honey@Yoga-7-2-in-1-14IML9:~/p7$ minikube service nginx-deployment
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:31281

```
🔗 Opening service default/nginx-deployment in default browser...
(base) 1rv24mc043_Honey@Yoga-7-2-in-1-14IML9:~/p7$ Created TensorFlow Lite XNNPACK delegate for CPU.
[11707:11736:1212/093256.209561:ERROR:google_apis/gcm/engine/registration_request.cc:292] Registration response error message: DEPRECATED_ENDPOINT
[11707:11736:1212/093323.047790:ERROR:google_apis/gcm/engine/registration_request.cc:292] Registration response error message: DEPRECATED_ENDPOINT
[11707:11736:1212/093409.062427:ERROR:google_apis/gcm/engine/registration_request.cc:292] Registration response error message: DEPRECATED_ENDPOINT
[11707:11736:1212/093539.393731:ERROR:google_apis/gcm/engine/registration_request.cc:292] Registration response error message: DEPRECATED_ENDPOINT
```

